



Find and Fix

Different programs, different bugs

Name:

Date:

★ I can find what kind of bug a program has and fix it.

Bugs come in different kinds

A bug can be a wrong block, an extra block, a missing block, or the wrong order. Read each program, run it, figure out the kind of bug, and fix it so Bit reaches the target.

Bit's number path

0

1

2

3

4

5

Program A — target is 4

start at 1

forward 1

1 Find and fix A

Run Program A from 1. Where does Bit land? It misses 4. Fix the program so Bit reaches 4 — what did you change?

Program B — target is 3

start at 0

forward 2

forward 1

forward 1

2 Find and fix B

Run Program B from 0. Where does Bit land? It misses 3. Fix the program so Bit reaches 3 — what did you change?

3 You try — name the bug

Each program had a different kind of bug. What kind of bug was in Program A? What kind was in Program B?

Big idea

A good debugger figures out WHAT kind of bug it is — wrong, extra, missing, or out of order — and then fixes that kind. Run the code again to make sure.



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Quick facilitation notes & answer key

What this worksheet teaches

Students diagnose and fix two programs with DIFFERENT kinds of bugs — a wrong block and an extra block — then name each bug type. Telling apart the kind of bug is a big step in debugging. No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil. You read each prompt aloud. Optional: tape a number line 0 to 5 on the floor and mark the target, so students can walk the buggy program and feel Bit miss.

How to lead it (about 20 minutes)

1. Show them (you do). Remind them of the bug types: a wrong block, an extra block, a missing block, a wrong-way block.
2. Do one together. Exercise 1 — Program A misses 4; the fix is to change a block's number.
3. Let them try. Students fix Program B (Ex 2), then name each bug type (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Program A: 1, forward 1 → 2 (misses 4). Bug = wrong block; change forward 1 to forward 3 so 1 → 4.

Exercise 2 — Program B: 0 → 2 → 3 → 4 (misses 3). Bug = an extra block; remove one forward 1 so 0 → 2 → 3.

Exercise 3 — Program A had a wrong block (wrong number); Program B had an extra block.

If students get stuck — and ways to adjust

Watch for: applying the same fix to both. Each program has a different kind of bug.

Make it easier: ask first "did Bit fall short, go too far, or go the wrong way?"

Make it harder: ask them to write a third program with a missing-block bug.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow (or write) step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read a program, change a block to fix it, and explain what the change did.

You'll know it worked when

The child fixes both programs and names the wrong-block and extra-block bug types.